2. Sputnik V vaccine is manufactured using vector based platform.
3. COVAXIN is an inactivated pathogen based vaccine.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (b) : Serum Institute's Covishield used the viral vector platform. It used a chimpanzee adenovirus, ChAdOx1, a type of common cold virus, which attaches itself to cells and injects DNA that tells the cells to make corona virus spike protein. Hence 1st statement is incorrect. Rests of the statements are correct.

40. If a major solar storm (solar-flare) reaches the Earth, which of the following are the possible effects on the Earth?

1. GPS and navigation systems could fail.
2. Tsunamis could occur at equatorial regions.
3. Power grids could be damaged.
4. Intense auroras could occur over much of the Earth.
5. Forest fires could take place over much of the planet.
6. Orbits of the satellites could be disturbed.
7. Shortwave radio communication of the aircraft flying over Polar Regions could be interrupted.

Select the correct answer using the code given below:

(a) 1, 2, 4 and 5 only (b) 2, 3, 5, 6 and 7 only
(c) 1, 3, 4, 6 and 7 only (d) 1, 2, 3, 4, 5, 6 and 7

Ans. (c) : Statements 2 and 5 are incorrect. Solar flares can temporarily alter the upper atmosphere creating disruptions. But it does not cause Tsunamis. It only affects the upper atmosphere. Also there is no causal link between the coronal mass ejections (CMEs) and the deadly forest fires. There is a common misconception that solar storms are associated with extreme geological and atmospheric effects on Earth.
Rests of the statements are correct.

41. "Climate Action Tracker" which monitors the emission reduction pledges of different countries is a:

(a) Database created by coalition of research organizations
(b) Wing of "International Panel of Climate Change"
(c) Committee under "United Nations Framework Convention on Climate Change"
(d) Agency promoted and financed by United Nations Environment Programme and World Bank

Ans. (a) : The Climate Action Tracker is an independent scientific analysis that tracks government climate action and measures it against the globally agreed Paris Agreement aim of "holding global warming well below 2°C, and pursuing efforts to limit warming to 1.5°C." A collaboration of two organisations, Climate Analytics and New Climate Institute, the Climate Action Tracker has been providing this independent analysis to policymakers since 2009.

42. Consider the following statements:

1. "The Climate Group" is an international non-profit organization that drives climate action by building large networks and runs them.
2. The International Energy Agency in partnership with the Climate Group launched a global initiative "EP100".
3. EP100 brings together leading companies committed to driving innovation in energy efficiency and increasing competitiveness while delivering on emission reduction goals.
4. Some Indian companies are members of EP100.
5. The International Energy Agency is the Secretariat to the "Under2 Coalition."

Which of the statements given above are correct?

(a) 1, 2, 4 and 5 (b) 1, 3 and 4 only
(c) 2, 3 and 5 only (d) 1, 2, 3, 4 and 5

Ans. (b) : EP100 is a global initiative led by The Climate Group and the Alliance to Save Energy, not by International Energy Agency. Hence 2nd statement is incorrect. Also, 5th Statement is incorrect as Climate Group is the Secretariat to the Under 2 Coalition and not the International Energy Agency. Rests of the statements are correct.

43. "If rainforests and tropical forests are the lungs of the Earth, then surely wetlands function as its kidneys." Which one of the following functions of wetlands best reflects the above statement?

(a) The water cycle in wetlands involves surface runoff, subsoil percolation and evaporation.
(b) Algae form the nutrient base upon which fish, crustaceans, molluscs, birds, reptiles and mammals thrive.
(c) Wetlands play a vital role in maintaining sedimentation balance and soil stabilization.
(d) Aquatic plants absorb heavy metals and excess nutrients

Ans. (d) : According to The Ramsar Convention, "wetlands are areas of marsh, fen, peatland or water, whether natural or artificial, permanent or temporary, with water that is static or flowing, fresh, brackish or salt, including areas of marine water, the depth of which at low tide does not exceed six metres."
Wetlands are vital for humans, for other ecosystems and for our climate, providing essential ecosystem services such as water regulation, including flood control and water purification. Wetlands also absorb carbon dioxide to help slow global heating and reduce pollution, hence have often been referred to as the "Kidneys of the Earth".
Therefore, option (d) is the correct answer.

44. In the context of WHO Air Quality Guidelines, consider the following statements:

1. The 24 hour mean of $PM_{2.5}$ should not exceed 15 $\mu g/m^3$ and annual mean of $PM_{2.5}$ should not exceed 5 $\mu g/m^3$.
2. In a year, the highest levels of ozone pollution occur during the periods of inclement weather.
3. PM_{10} can penetrate the lung barrier and enter the bloodstream.
4. Excessive ozone in the air can trigger asthma.

Which of the statements given above are correct?

(a) 1, 3 and 4 (b) 1 and 4 only
(c) 2, 3 and 4 (d) 1 and 2 only

Ans. (b) : Ozone at ground level is formed by the reaction with sunlight (photochemical reaction) of pollutants such as nitrogen oxides (NOx) from vehicle and industry emissions and volatile organic compounds (VOCs) emitted by vehicles, solvents and industry. As a result, the highest levels of ozone pollution occur during periods of sunny weather and not during inclement weather. Hence 2nd statement is incorrect. Again, while particles with a diameter of 10 microns or less, (≤ PM10) can penetrate and lodge deep inside the lungs, the even more health-damaging particles are those with a diameter of 2.5 microns or less, (≤ PM2.5). PM2.5 can penetrate the lung barrier and enter the blood system. Hence 3rd statement is also incorrect.
Rests of the statements are correct.

45. With reference to "Gucchi" sometimes mentioned in the news, consider the following statements:

1. It is a fungus.
2. It grows in some Himalayan forest areas.
3. It is commercially cultivated in the Himalayan foothills of north-eastern India.

Which of the statements given above is/are correct?

(a) 1 only (b) 3 only (c) 1 and 2 (d) 2 and 3

Ans. (c) : Guchhi mushroom is a species of fungus in the family Morchellaceae of the Ascomycota. They are pale yellow in colour with large pits and ridges on the surface of the cap, raised on a large white stem. guchhi faces lots of challenges in its cultivation. The mushrooms cannot be cultivated commercially and grow in conifer forests across temperature regions, and the foothills in Himachal Pradesh, Uttarakhand, and Jammu and Kashmir. And it takes months for villagers to collect enough of these mushrooms, dry them and bring them to the market.

46. With reference to polyethylene terephthalate, the use of which is so widespread in our daily lives, consider the following statements:

1. Its fibres can be blended with wool and cotton fibres to reinforce their properties.
2. Containers made of it can be used to store any alcoholic beverage.
3. Bottles made of it can be recycled into other products.
4. Articles made of it can be easily disposed of by incineration without causing greenhouse gas emissions.

Which of the statement given above are correct?

(a) 1 and 3 (b) 2 and 4 (c) 1 and 4 (d) 2 and 3

Ans. (a) : Plastic packaging is dangerous to human health and the country liquor and country made foreign liquor cannot be sold in such bottles. That's why Maharashtra government has announced the ban on sale of alcohol in PET bottles from April 1. Therefore 2nd statement is incorrect. Again, Polyethylene, like other plastics, is not inert and is known to release additives and other degradation products into the environment throughout its lifetime. Hence 4th statement is also incorrect.
PET fibres are often used in durable-press blends with other fibres such as rayon, wool, and cotton, reinforcing the inherent properties of those fibres while contributing to the ability of the fabric to recover from wrinkling. Also Polyethylene terephthalate is the most widely recycled plastic.

47. Which of the following is not a bird?

(a) Golden Mahseer (b) Indian Nightjar
(c) Spoonbill (d) White Ibis

Ans. (a): Golden Mahseer is a fish that lives in fast-moving waters, inhabiting hill streams with a rocky and stony substrate. Rests are type of birds.

48. Which of the following are nitrogen-fixing plants?

1. Alfalfa 2. Amaranth
3. Chickpea 4. Clover
5. Purslane (Kulfa) 6. Spinach

Select the correct answer using the code given below:

(a) 1, 3 and 4 only (b) 1, 3, 5 and 6 only
(c) 2, 4, 5 and 6 only (d) 1, 2, 4, 5 and 6

Ans. (a): Out of the given options only Alfalfa, Chickpea and Clovers are nitrogen- fixing plants. Nitrogen-Fixing Plants are capable of fixing Nitrogen in a natural way to provide plant-adjusted Nitrogen without any industrial harm to nature.

49. "Biorock technology" is talked about in which one of the following situations?

(a) Restoration of damaged coral reefs
(b) Development of building materials using plant residues
(c) Identification of areas for exploration/ extraction of shale gas
(d) Providing salt licks for wild animals in forest/protected areas

Ans. (a) : The Zoological Survey of India (ZSI), with help from Gujarat's forest department, is attempting a process for the first time to restore coral reefs using bio rock or mineral accretion technology. A bio rock structure was installed one nautical mile off the Mithapur coast in the Gulf of Kachchh.

50. The "Miyawaki method' is well known for the:

(a) Promotion of commercial farming in arid and semi-arid areas
(b) Development of gardens using genetically modified flora
(c) Creation of mini forests in urban areas
(d) Harvesting wind energy on coastal area and on sea surfaces.

Ans. (c) : Miyawaki is a method of afforestation that requires quite a small space, at least 20 square feet. One must seed plants very close to save space and dense plant growth. It was developed by Akira Miyawaki of Japan. Bombay Municipal Corporation (BMC) has been using the Miyawaki method to create tiny urban forests in the Metropolitan areas of Mumbai.

51. In the Government of India Act 1919, the functions of Provincial Government were divided into "Reserved" and "Transferred" subjects. Which of the following were treated as "Reserved" subjects ?

1. Administration of Justice
2. Local Self-Government
3. Land Revenue
4. Police

Select the correct answer using the code given below

(a) 1, 2 and 3 (b) 2, 3 and 4 (c) 1, 3 and 4 (d) 1, 2 and 4

Ans. (c): Subjects were divided into two lists reserved and transferred in the Government of India act 1919.
Reserved subjects: Law and order, finance, land revenue, irrigation, police etc. to be administered by the governor through his executive council.
Transferred subjects: Education, health, local government, industry, agriculture, excise etc. to be administered by ministers nominated from among the elected members of legislative council.

52. In medieval India, the term "Fanam" referred to:

(a) Clothing (b) Coins (c) Ornaments (d) Weapons

Ans. (b): The Fanam coins were the regular unit of currency in medieval Travancore and appear to have been extensively used for trading in the region of South India. The words Fanam and Panam literally mean money and are still used as a synonym for wealth in Kerala in the native language of Malayalam.